NETWORKING / NET03 / NOVEMBER 2026

Access Switch Replacement, Migration & Rollback

A planned network change in Cisco Packet Tracer

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The objective

I replaced the access switch serving Sales, HR, and Marketing in my NET01 office network. The goal was to preserve VLAN membership, DHCP, and routed connectivity while following a clear port map and keeping a working way back to the original switch.

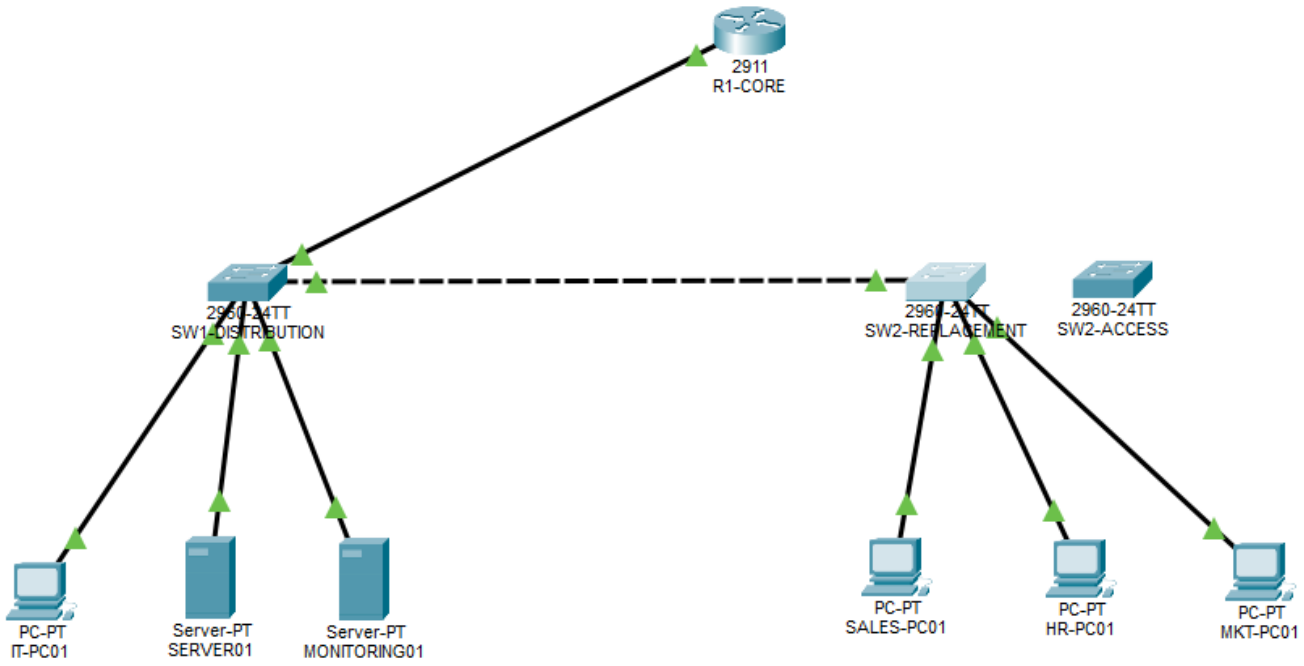


Figure 1. Final state: SW2-REPLACEMENT serves the three departments. The original SW2-ACCESS remains uncabled as the rollback spare.

3 departments	4 connections	DHCP verified	Rollback tested
Sales, HR, Marketing	One uplink, three PCs	Release and renew	Original switch restored

Result

All three PCs reached their gateways and SERVER01 after the first cutover, after rollback, and after the final cutover. DHCP renewal succeeded through the replacement. The final six gateway/server ping tests each completed with four replies and 0% loss.

This is a guided personal lab using simulated Cisco switches. It demonstrates a planned replacement and recovery procedure, not a production deployment, an automatic failover, or a zero-downtime migration.

01 / PRE-CHANGE SURVEY

Record what the replacement must preserve

I surveyed the original switch before changing cables. Interface status identified three connected departmental ports. VLAN membership and trunk output established the required Layer 2 configuration; CDP confirmed the upstream connection.

```
SW2-ACCESS#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/2
10   IT                               active
20   SALES                            active    Fa0/1
30   HR                               active    Fa0/2
40   MARKETING                        active    Fa0/3
110  SERVERS                          active
160  MONITORING                       active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW2-ACCESS#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      20,30,40

Port        Vlans allowed and active in management domain
Gig0/1      20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      20,30,40
```

Figure 2. Original SW2: access VLANs 20, 30, and 40; Gi0/1 carries those same VLANs and shows them forwarding.

```
SW2-ACCESS#show cdp ne
SW2-ACCESS#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce   Holdtme    Capability   Platform    Port ID
SW1-DISTRIBUTION
             Gig 0/1          130            S       2960        Gig 0/2
```

Figure 3. CDP confirms local SW2 Gi0/1 connects to SW1-DISTRIBUTION Gi0/2.

Connection	Original port	Replacement port	Role
SALES-PC01 Fa0	Fa0/1	Fa0/1	Access VLAN 20
HR-PC01 Fa0	Fa0/2	Fa0/2	Access VLAN 30
MKT-PC01 Fa0	Fa0/3	Fa0/3	Access VLAN 40
SW1 Gi0/2	Gi0/1	Gi0/1	Trunk 20,30,40

I retained the same port numbering to make the cutover and rollback easy to check. The original switch also contained VLANs 10, 110, and 160. I preserved that database on the replacement, while allowing only VLANs 20, 30, and 40 on this uplink. Native VLAN 1 and PVST were retained.

02 / BASELINE & CHANGE PLAN

Define success before disconnecting anything

Before the migration, I checked each PC's IP settings, its own gateway, and SERVER01 at 10.10.20.100. These same destinations became the acceptance checks for the replacement and for rollback.

Client	Observed address /24	Gateway	Gateway / server
Sales	10.20.20.100	10.20.20.1	Both pass
HR	10.20.30.100	10.20.30.1	Both pass
Marketing	10.20.40.100	10.20.40.1	Both pass

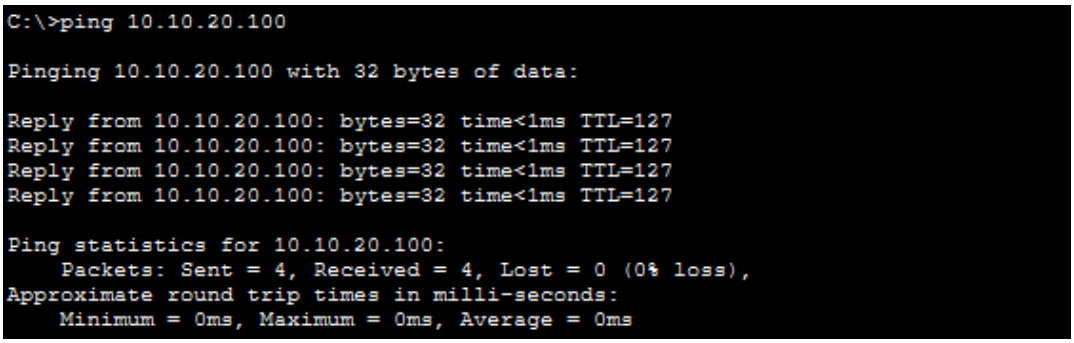


Figure 4. Sales baseline server test: the repeat received all four replies. HR and Marketing also passed their baseline tests.

Sales lost the first reply in its initial server test. I repeated the test and obtained 0% loss before starting the change. I did not treat that single initial timeout as evidence that switch configuration needed changing.

The change sequence

Stage	Action and decision
Prepare	Record VLANs, access ports, uplink, and working client paths. Retain the original switch configuration.
Stage	Configure a separate, uncabled 2960 replacement. Verify access membership and administrative trunk settings, then save.
Cut over	Move the uplink first and verify it. Move the three PC connections to their mapped ports, then repeat client tests.
Accept	Check all six gateway/server paths and obtain DHCP leases again through the replacement.
Rehearse recovery	Move the same four connections back to the original switch and verify recovery; then repeat the cutover for the final state.

The three departmental PCs lose upstream connectivity during the cable moves. No interruption duration was measured. The recovery action for unresolved connectivity or DHCP failure would be to reconnect the original switch.

03 / REPLACEMENT STAGING

Prepare the switch while the office network stays connected

I placed another 2960-24TT beside SW2 and left it uncabled. I set the hostname to SW2-REPLACEMENT, matched PVST and the six VLAN names, and configured Fa0/1-Fa0/3 as explicit access ports in VLANs 20, 30, and 40.

```
SW2-REPLACEMENT#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
10   IT                               active
20   SALES                            active    Fa0/1
30   HR                               active    Fa0/2
40   MARKETING                        active    Fa0/3
110  SERVERS                          active
160  MONITORING                       active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Figure 5. The replacement VLAN table matches the planned departmental access-port membership.

For Gi0/1, I configured trunk mode with the allowed list 20,30,40. The pre-cutover check showed administrative trunk mode, native VLAN 1, and the intended allowed list. Operational mode was down because the switch was still uncabled.

```
SW2-REPLACEMENT#show interfaces G
SW2-REPLACEMENT#show interfaces GigabitEthernet0/1 switchport
Name: Gig0/1
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: down
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: 20,30,40
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none
```

Figure 6. Administrative trunk settings are present before the physical link is connected.

```
SW2-REPLACEMENT#copu r
SW2-REPLACEMENT#copy r
SW2-REPLACEMENT#copy running-config st
SW2-REPLACEMENT#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
SW2-REPLACEMENT#
```

Figure 7. I added no ip domain-lookup and saved the staged configuration; the save returned [OK].

Two mistyped commands were rejected during entry: a VLAN number typo and switchport trunk allows vlan. I corrected them and checked the resulting state. Rejected commands did not establish the intended configuration; verification output did.

04 / FIRST CUTOVER

Verify the uplink before moving the users

I disconnected the original SW2 uplink and connected SW1 Gi0/2 to replacement Gi0/1. The original switch kept the three PC connections during this checkpoint. After spanning-tree convergence, I checked trunk forwarding and the CDP neighbor.

```
SW2-REPLACEMENT#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      20,30,40

Port        Vlans allowed and active in management domain
Gig0/1      20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      20,30,40

SW2-REPLACEMENT#show cdp ne
SW2-REPLACEMENT#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce   Holdtme    Capability   Platform    Port ID
SW1-DISTRIBUTION
             Gig 0/1          142            S       2960        Gig 0/2
SW2-REPLACEMENT#
```

Figure 8. Replacement Gi0/1 is trunking and forwarding VLANs 20,30,40; CDP confirms SW1 on remote Gi0/2.

I then moved Sales, HR, and Marketing to Fa0/1, Fa0/2, and Fa0/3 on the replacement. Switch-to-switch links used copper cross-over; PC links used copper straight-through in Packet Tracer.

```
SW2-REPLACEMENT#show interfaces st
SW2-REPLACEMENT#show interfaces status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1     SALES-PC01         connected    20         a-full  a-100 10/100BaseTX
Fa0/2     HR-PC01            connected    30         a-full  a-100 10/100BaseTX
Fa0/3     MKT-PC01           connected    40         a-full  a-100 10/100BaseTX
```

Figure 9. The three access ports are connected in their intended VLANs after the cable moves.

Client acceptance checks

Client	Gateway	SERVER01	Result after first cutover
Sales	10.20.20.1	10.10.20.100	4/4 replies on both tests
HR	10.20.30.1	10.10.20.100	4/4 replies on both tests
Marketing	10.20.40.1	10.10.20.100	4/4 replies on both tests

Link lights alone were not the acceptance test. The gateway and server checks confirmed that each access VLAN could cross the new trunk and reach the router and the server subnet.

DHCP needs a separate check

The PCs could keep their existing leases during a cable move. Successful pings would therefore not establish that DHCP still worked through the replacement. I deliberately released and renewed the three leases next.

05 / DHCP VALIDATION

Test address acquisition, not just retained addresses

On each department PC, I ran ipconfig /release, ipconfig /renew, and ipconfig. Each release cleared the IPv4 settings; each renewal returned an address, /24 mask, and the correct department gateway through the replacement switch.

```
C:\>ipconfig /release

   IP Address.......................: 0.0.0.0
   Subnet Mask......................: 0.0.0.0
   Default Gateway..................: 0.0.0.0
   DNS Server.......................: 0.0.0.0

C:\>ipconfig /renew

   IP Address.......................: 10.20.20.100
   Subnet Mask......................: 255.255.255.0
   Default Gateway..................: 10.20.20.1
   DNS Server.......................: 0.0.0.0
```

Figure 10. Sales releases its address and obtains 10.20.20.100/24 with gateway 10.20.20.1.

```
C:\>ipconfig /release

   IP Address.......................: 0.0.0.0
   Subnet Mask......................: 0.0.0.0
   Default Gateway..................: 0.0.0.0
   DNS Server.......................: 0.0.0.0

C:\>ipconfig /renew

   IP Address.......................: 10.20.30.100
   Subnet Mask......................: 255.255.255.0
   Default Gateway..................: 10.20.30.1
   DNS Server.......................: 0.0.0.0
```

Figure 11. HR obtains 10.20.30.100/24 with gateway 10.20.30.1.

```
C:\>ipconfig /release

   IP Address.......................: 0.0.0.0
   Subnet Mask......................: 0.0.0.0
   Default Gateway..................: 0.0.0.0
   DNS Server.......................: 0.0.0.0

C:\>ipconfig /renew

   IP Address.......................: 10.20.40.100
   Subnet Mask......................: 255.255.255.0
   Default Gateway..................: 10.20.40.1
   DNS Server.......................: 0.0.0.0
```

Figure 12. Marketing obtains 10.20.40.100/24 with gateway 10.20.40.1.

After renewal, all three PCs again reached SERVER01 with four replies and 0% loss. R1 remained the DHCP server and router; I did not move DHCP onto the switch or alter the router pools.

Why this mattered

Releasing the leases tested address acquisition independently of the addresses already held. DNS remained unset, as in NET01; name resolution was outside this change.

06 / ROLLBACK REHEARSAL

Prove that the original switch can restore service

After the first cutover passed, I intentionally moved the uplink and all three PC connections back to SW2-ACCESS. The replacement stayed configured but uncabled. This was a planned recovery rehearsal, not a response to a failed replacement.

```
SW2-ACCESS#show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      20,30,40

Port        Vlans allowed and active in management domain
Gig0/1      20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      20,30,40

SW2-ACCESS#
```

Figure 13. Original SW2 Gi0/1 is trunking again, with VLANs 20,30,40 allowed, active, and forwarding.

All three gateway checks passed. The server results below confirm routed reachability after restoring the original switch.

```
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 14. Rollback: Sales to SERVER01, four replies and 0% loss.

```
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 15. Rollback: HR to SERVER01, four replies and 0% loss.

```
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 16. Rollback: Marketing to SERVER01, four replies and 0% loss.

07 / FINAL CUTOVER

Return to the replacement and repeat the acceptance checks

With rollback verified, I moved the same four connections back to SW2-REPLACEMENT. The original switch remained uncabled as a spare. The final topology is shown on the first page.

```
SW2-REPLACEMENT#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      20,30,40

Port        Vlans allowed and active in management domain
Gig0/1      20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      20,30,40

SW2-REPLACEMENT#
```

Figure 17. Final replacement trunk: VLANs 20,30,40 are allowed, active, and forwarding; native VLAN is 1.

Each department again reached its own gateway with 0% loss. These final server tests completed successfully as well.

```
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 18. Final state: Sales to SERVER01, four replies and 0% loss.

```
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 19. Final state: HR to SERVER01, four replies and 0% loss.

```
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 20. Final state: Marketing to SERVER01, four replies and 0% loss.

At closeout I confirmed completion of the final configuration save and the Packet Tracer file save. A successful staged configuration save is captured earlier in this report. I did not perform a switch reload to test persistence.

08 / HANDOFF & LESSONS

Treat a replacement as a controlled change

What the project demonstrates

I surveyed an existing Cisco access switch, staged a replacement, migrated four connections, checked forwarding and client reachability, verified DHCP acquisition, rehearsed rollback, and returned to the replacement for the final state. The work extends NET01 from configuration practice into a complete planned change.

Validation record

Checkpoint	Observed result
Pre-change baseline	All three departments reached their own gateway and SERVER01; the repeated Sales server test had 0% loss.
First cutover	Correct connected access VLANs, forwarding trunk, expected CDP neighbor, and six successful client ping tests.
DHCP acquisition	All three release/renew tests succeeded in the intended subnets; all three post-renewal server pings had 0% loss.
Rollback	Original trunk forwarding restored; all six gateway/server checks passed.
Final cutover	Replacement trunk forwarding restored; all six gateway/server checks passed with 0% loss.

Lessons I would use in network support

Record the upstream port and endpoint map before moving cables. Stage and inspect the replacement before taking the users offline. Check administrative settings while uncabled, then operational forwarding after connection. Keep the original switch ready, and verify recovery from the PCs as well as from the switch CLI.

Scope and limits

This was a same-model Cisco Packet Tracer replacement with a planned service interruption. It was not a cross-vendor migration, firmware upgrade, or production maintenance event. No outage duration, load performance, application availability, all-pairs reachability, or reboot persistence was tested. Gateway pings and server pings validate the specific ICMP paths shown.

Project files and closeout

The companion change-reference.md records the port map, staged configuration, verification commands, and reverse cable map. It is reconstructed from the documented session, not an exported running configuration. The completed .pkt file was saved locally by the lab operator and was not supplied for this publication.

Portfolio: carlostibj.github.io/project-net03.html
Contact: carlomtj15@gmail.com